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- Posterior distributions
- Predictive distributions
- Prior distributions
- The Normal model
- Poisson Model

Bayesian Statistics and Data Analysis

Lecture 2

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Thanks to Aki Vehtari, Aalto University



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Binomial: known θ

- Posterior distributions
- Predictive distributions
- Prior distributions
- The Normal model
- Poisson Model

- Probability of event 1 in trial is θ



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Binomial: known θ

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- Probability of event 1 in trial is θ
- Probability of event 2 in trial is $1 - \theta$



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- Probability of event 1 in trial is θ
- Probability of event 2 in trial is $1 - \theta$
- Probability of several events in independent trials is e.g.
 $\theta\theta(1 - \theta)\theta(1 - \theta)(1 - \theta)\dots$



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- Probability of event 1 in trial is θ
- Probability of event 2 in trial is $1 - \theta$
- Probability of several events in independent trials is e.g. $\theta\theta(1 - \theta)\theta(1 - \theta)(1 - \theta)\dots$
- If there are n trials and we don't care about the order of the events, then the probability that event 1 happens y times is

$$p(y|\theta, n) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$$



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- Observation model (function of y , discrete)

$$p(y|\theta, n) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$$



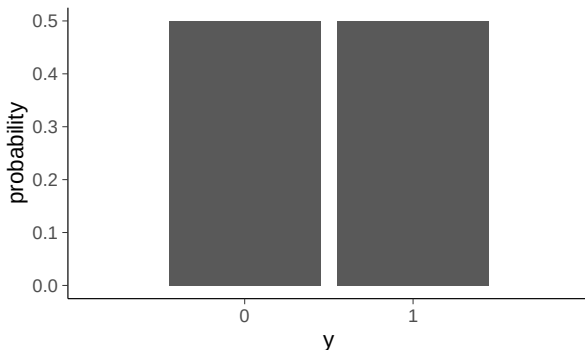
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Binomial: known θ

- Observation model (function of y , discrete)

$$p(y|\theta, n) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$$

Binomial distribution with $\theta = 0.5$, $n=1$





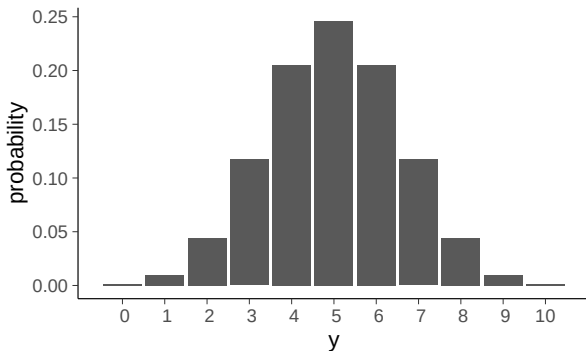
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$$p(y|\theta, n) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$$

Binomial distribution with $\theta = 0.5$, $n=10$





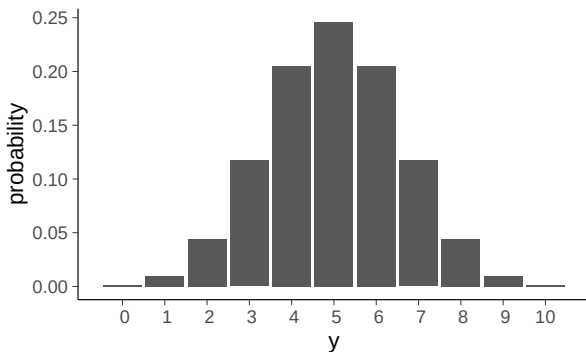
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Binomial: known θ

- Observation model (function of y , discrete)

$$p(y|\theta, n) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$$

Binomial distribution with $\theta = 0.5$, $n=10$



$p(y|n = 10, \theta = 0.5)$: 0.00 0.01 0.04 0.12 0.21 0.25 0.21 0.12 0.04 0.01 0.00

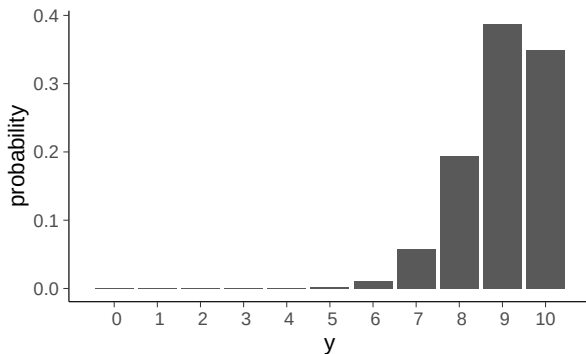


Binomial: known θ

- Observation model (function of y , discrete)

$$p(y|\theta, n) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$$

Binomial distribution with $\theta = 0.9$, $n=10$



$p(y|n = 10, \theta = 0.9)$: 0.00 0.00 0.00 0.00 0.00 0.00 0.01 0.06 0.19 0.39 0.35



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Binomial: unknown θ

- Posterior with Bayes rule (function of θ , continuous)

$$p(\theta|y, n, M) = \frac{p(y|\theta, n, M)p(\theta|n, M)}{p(y|n, M)}$$

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- Start with uniform prior

$$p(\theta|n, M) = p(\theta|M) = 1, \text{ when } 0 \leq \theta \leq 1$$



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- Start with uniform prior

$$p(\theta|n, M) = p(\theta|M) = 1, \text{ when } 0 \leq \theta \leq 1$$

- **Discussion question:** Which parts of this expression change when θ changes, and which parts are fixed after observing y and n ?



$$p(\theta|y, n, M) = \frac{p(y|\theta, n, M)p(\theta|M)}{p(y|n, M)} = \frac{p(y|\theta, n, M)}{p(y|n, M)}$$

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- Uniform prior: $p(\theta|M) = 1$ on $0 \leq \theta \leq 1$
- The prior appears in the numerator and denominator
- Since it is 1, it drops out



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Binomial: unknown θ

$$\begin{aligned} p(\theta|y, n, M) &= \frac{p(y|\theta, n, M)p(\theta|M)}{p(y|n, M)} = \frac{p(y|\theta, n, M)}{p(y|n, M)} \\ &= \frac{\binom{n}{y} \theta^y (1-\theta)^{n-y}}{p(y|n, M)} \end{aligned}$$

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- Uniform prior: $p(\theta|M) = 1$ on $0 \leq \theta \leq 1$
- The prior appears in the numerator and denominator
- Since it is 1, it drops out
- Z is constant as a function of θ



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- Uniform prior: $p(\theta|M) = 1$ on $0 \leq \theta \leq 1$
- The prior appears in the numerator and denominator
- Since it is 1, it drops out
- Z is constant as a function of θ
- For inference, keep the part that depends on θ



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Binomial: unknown θ

- Normalization term Z (constant given y)

$$Z = \int_0^1 \theta^y (1 - \theta)^{n-y} d\theta = B(y + 1, n - y + 1)$$

- Normalisation term has *Beta* function form

$$B(a, b) = \int_0^1 t^{a-1} (1 - t)^{b-1} dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

- Where does $B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$ come from?
- with integers $\Gamma(n) = (n-1)!$
(for large integers even this is challenging and usually $\log \Gamma(\cdot)$ is computed instead of $\Gamma(\cdot)$)
- **Study recommendation:** ask an LLM to step through the standard-integral derivation, or work it through yourself using change of variables



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- Posterior is

$$p(\theta|y, n, M) = \frac{\Gamma(n+2)}{\Gamma(y+1)\Gamma(n-y+1)} \theta^y (1-\theta)^{n-y},$$



• Posterior distributions

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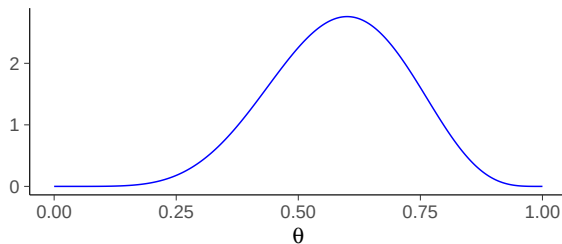
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$$p(\theta|y, n, M) = \frac{\Gamma(n+2)}{\Gamma(y+1)\Gamma(n-y+1)}\theta^y(1-\theta)^{n-y},$$

which is called Beta distribution

$$\theta|y, n \sim \text{Beta}(y+1, n-y+1)$$

$p(\theta | y=6, n=10, M=\text{binom}) + \text{unif. prior}$





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Binomial: computation

- R
 - density `dbeta`
 - CDF `pbeta`
 - quantile `qbeta`
 - random number `rbeta`
- Python
 - `from scipy.stats import beta`
 - density `beta.pdf`
 - CDF `beta.cdf`
 - prctile `beta.ppf`
 - random number `beta.rvs`
- CDFs and quantiles are numerically nontrivial, so use library functions
- **Study recommendation:** use an LLM to dig deeper into why CDFs and quantiles often require numerical methods



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Placenta previa

- Probability of a girl birth given placenta previa (BDA3 p. 37)
 - 437 girls and 543 boys have been observed
 - is the ratio 0.445 different from the population average 0.485?

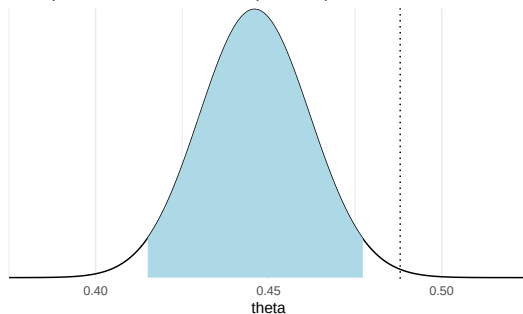


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- Probability of a girl birth given placenta previa (BDA3 p. 37)
 - 437 girls and 543 boys have been observed
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Uniform prior $\rightarrow$ Posterior is Beta(438,544)



95% posterior interval



Predictive distribution – Effect of integration

- **Discussion:** Why integrate over θ instead of plug in one estimate?

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Predictive distribution – Effect of integration

- **Discussion:** Why integrate over θ instead of plug in one estimate?
- Predictive distribution for new $\tilde{y}$ (discrete)

$$p(\tilde{y} = 1|y, n, M) = \int_0^1 p(\tilde{y} = 1|\theta, y, n, M)p(\theta|y, n, M)d\theta$$

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- With uniform prior

$$E[\theta|y] = \frac{y+1}{n+2}$$



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- With uniform prior

$$E[\theta|y] = \frac{y+1}{n+2}$$

- Extreme cases

$$\begin{aligned}p(\tilde{y} = 1|y = 0, n, M) &= \frac{1}{n+2} \\p(\tilde{y} = 1|y = n, n, M) &= \frac{n+1}{n+2}\end{aligned}$$



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- Maximum likelihood estimate

$$\hat{\theta}_{\text{MLE}} = \frac{y}{n}$$

- Posterior mean with uniform prior

$$E[\theta|y] = \frac{y+1}{n+2}$$

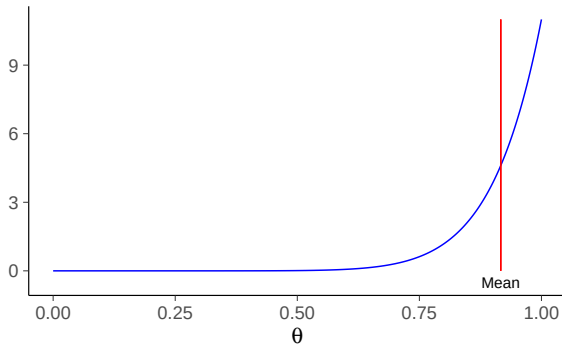
- Discussion question: Why can the posterior mean be a better estimate than the maximum likelihood estimate when n is small?



Benefits of integration

Example: $n = 10, y = 10$

Posterior of θ of Binomial model with $y=10, n=$





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- Prior predictive distribution for new $\tilde{y}$ (discrete)

$$p(\tilde{y} = 1|M) = \int_0^1 p(\tilde{y} = 1|\theta, y, n, M)p(\theta|M)d\theta$$

- Posterior predictive distribution for new $\tilde{y}$ (discrete)

$$p(\tilde{y} = 1|y, n, M) = \int_0^1 p(\tilde{y} = 1|\theta, y, n, M)p(\theta|y, n, M)d\theta$$



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- $p(\theta|M) = 1$ if
 - 1) we want the prior predictive distribution to be uniform

$$p(\tilde{y} = 1 | n = 0, M) = \frac{1}{2}$$

- nice justification as it is based on observables y and n



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- nice justification as it is based on observables y and n

2) we think all values of θ are equally likely



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Priors

- Posterior distributions
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- Conjugate prior (BDA3 p. 35)
- Noninformative prior (BDA3 p. 51)
- Proper and improper prior (BDA3 p. 52)
- Weakly informative prior (BDA3 p. 55)
- Informative prior (BDA3 p. 55)
- Prior sensitivity (BDA3 p. 38)



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Conjugate priors

- A prior is *conjugate* for a likelihood if the posterior is in the same distribution family as the prior

prior family + likelihood $\Rightarrow$ same posterior family



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- Example:

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$y|\theta \sim \text{Binomial}(n, \theta)$$

$$\theta|y \sim \text{Beta}(\alpha + y, \beta + n - y)$$



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- Why useful?
 - gives closed-form posteriors
 - helps us understand Bayesian updating
 - provides checks for simulation-based methods
 - can sometimes allow analytic marginalization



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- Why useful?
 - gives closed-form posteriors
 - helps us understand Bayesian updating
 - provides checks for simulation-based methods
 - can sometimes allow analytic marginalization
- But conjugacy is not required: later we use simulation and Stan when closed forms are unavailable



- Posterior distributions
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Beta prior for Binomial model

- Prior

$$\text{Beta}(\theta|\alpha, \beta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

- Posterior

$$p(\theta|y, n, M) \propto \theta^y(1-\theta)^{n-y}\theta^{\alpha-1}(1-\theta)^{\beta-1}$$



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after normalization

$$p(\theta|y, n, M) = \text{Beta}(\theta|\alpha + y, \beta + n - y)$$



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- **Discussion question:** What do α and β behave like in terms of prior successes/failures?



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- **Discussion question:** What do α and β behave like in terms of prior successes/failures?
- α and β can be considered to be number of prior observations



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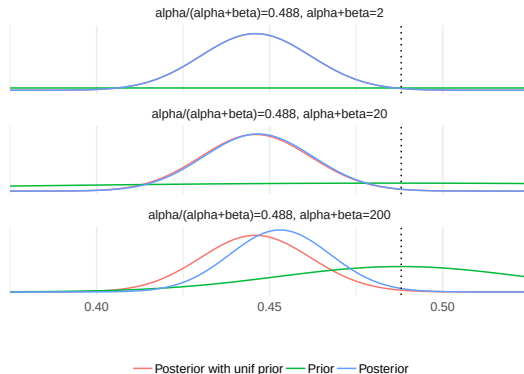
$$p(\theta|y, n, M) = \text{Beta}(\theta|\alpha + y, \beta + n - y)$$

- **Discussion question:** What do α and β behave like in terms of prior successes/failures?
- α and β can be considered to be number of prior observations
- Uniform prior when $\alpha = 1$ and $\beta = 1$



Placenta previa

- Beta prior centered on population average 0.485



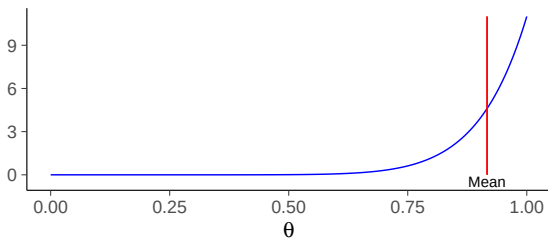


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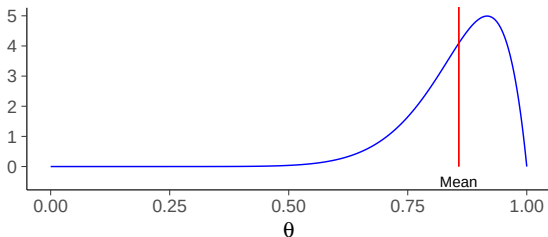
Benefits of integration and prior

Example: $n = 10, y = 10$ - uniform vs Beta(2,2) prior

$p(\theta | y=10, n=10, M=\text{binom}) + \text{unif. prior}$



$p(\theta | y=10, n=10, M=\text{binom}) + \text{Beta}(2,2) \text{ prior}$





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Beta prior for Binomial model

- Posterior

$$p(\theta|y, n, M) = \text{Beta}(\theta|\alpha + y, \beta + n - y)$$

- Posterior mean

$$E[\theta|y] = \frac{\alpha + y}{\alpha + \beta + n}$$

- combination prior and likelihood information
- when $n \rightarrow \infty$, $E[\theta|y] \rightarrow y/n$
- **Discussion question:** Why can the posterior mean be a better estimate than the maximum likelihood estimate y/n when n is small?



- Posterior distributions
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Beta prior for Binomial model

- Posterior

$$p(\theta|y, n, M) = \text{Beta}(\theta|\alpha + y, \beta + n - y)$$

- Posterior mean

$$E[\theta|y] = \frac{\alpha + y}{\alpha + \beta + n}$$

- combination prior and likelihood information
- when $n \rightarrow \infty$, $E[\theta|y] \rightarrow y/n$
- **Discussion question:** Why can the posterior mean be a better estimate than the maximum likelihood estimate y/n when n is small?
- Posterior variance

$$\text{var}[\theta|y] = \frac{E[\theta|y](1 - E[\theta|y])}{\alpha + \beta + n + 1}$$

- decreases when n increases
- when $n \rightarrow \infty$, $\text{var}[\theta|y] \rightarrow 0$



Weakly informative and “noninformative” priors

- Posterior distributions
- Predictive distributions
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- Poisson Model

- Goal: use a prior that does not dominate the likelihood
- Common terms: vague, flat, diffuse, weakly informative



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- But “flat” does not always mean “noninformative”
 - flat in θ is not flat in $\log(\theta)$
 - flat in θ is not flat in $\text{logit}(\theta)$



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- Goal: use a prior that does not dominate the likelihood
- Common terms: vague, flat, diffuse, weakly informative
- But “flat” does not always mean “noninformative”
 - flat in θ is not flat in $\log(\theta)$
 - flat in θ is not flat in $\text{logit}(\theta)$
- Prior choice should be checked on the scale that matters
- Prefer weakly informative priors when possible: rule out unreasonable values without forcing a narrow answer



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Example of informative prior

- Posterior distributions
- Predictive distributions
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- Poisson Model

- The percentage of girl births is remarkably stable at about 48.8% girls, rarely varying by more than 0.5% from this rate



Example of informative prior

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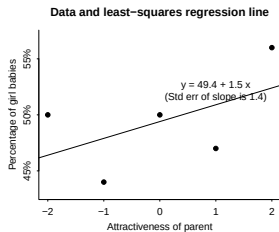
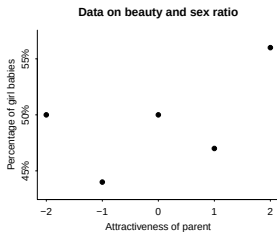
- The percentage of girl births is remarkably stable at about 48.8% girls, rarely varying by more than 0.5% from this rate
- There was a study on the percentage of girl births among parents in attractiveness categories 1–5 (assessed by interviewers in a face-to-face survey)



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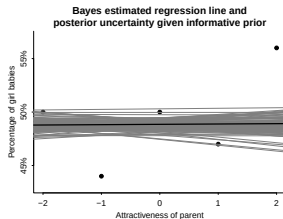
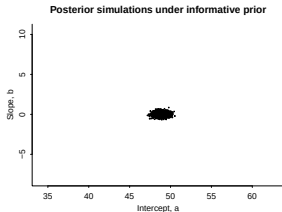
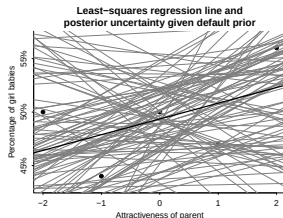
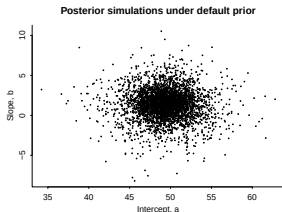




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Proper and improper priors

- A proper prior integrates to one:

$$\int p(\theta) d\theta = 1$$

- An improper prior does not have a finite integral:

$$\int p(\theta) d\theta = \infty$$



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- Example: binomial model with $p(\theta) \propto \theta^{-1}(1 - \theta)^{-1}$

$$\theta|y, n \sim \text{Beta}(y, n - y)$$

- This posterior is proper only if $0 < y < n$



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- This posterior is proper only if $0 < y < n$
- Always check that the posterior is proper



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Weakly informative priors

- Posterior distributions
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- Weakly informative priors produce computationally better behaving posteriors
 - If we want to model IQ in children, how to construct a prior?



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Weakly informative priors

- Weakly informative priors produce computationally better behaving posteriors
 - If we want to model IQ in children, how to construct a prior?
 - often there's some knowledge about the scale
 - Using the **prior predictive** distribution

$$p(\tilde{y}|M) = \int p(\tilde{y}|\theta, M)p(\theta|M)d\theta,$$

we can simulate data from the model:

Does it look (remotely) reasonable?

- useful if there's more information from previous observations - not certain how well that information is applicable in a new case



Construction of weakly informative priors

- Posterior distributions
- Predictive distributions
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- Prior prediction checks!
- Start with some version of a noninformative prior, then add information until reasonable.
- Start with a strong prior, then broaden it to account for uncertainty



Construction of weakly informative priors

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- Prior prediction checks!
- Start with some version of a noninformative prior, then add information until reasonable.
- Start with a strong prior, then broaden it to account for uncertainty
- Stan team prior choice recommendations
<https://github.com/stan-dev/stan/wiki/Prior-Choice-Recommendations>



Example of weakly informative prior

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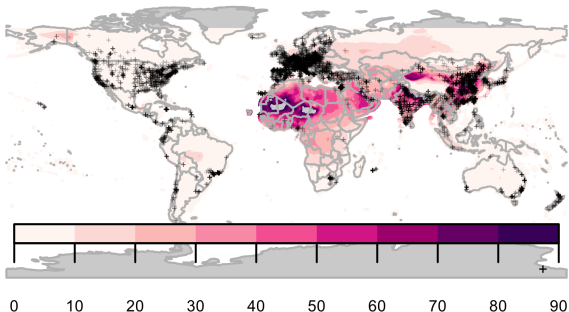
- Gabry et al (2019). Visualization in Bayesian workflow.
 - Estimation of human exposure to air pollution from particulate matter measuring less than 2.5 microns in diameter ($PM_{2.5}$)
 - A recent report estimated that $PM_{2.5}$ is responsible for three million deaths worldwide each year (Shaddick et al, 2017)



Example of weakly informative prior

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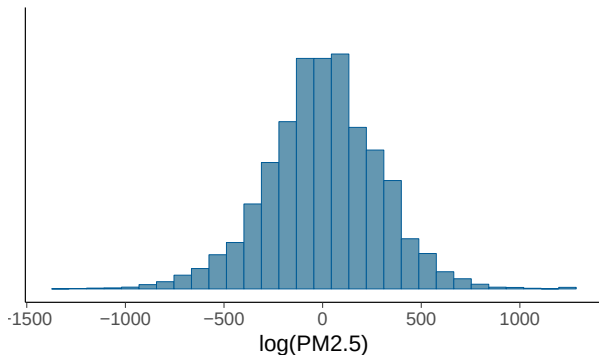


Satellite estimates and ground monitor locations



Example of weakly informative prior

Prior predictive distribution with vague prior

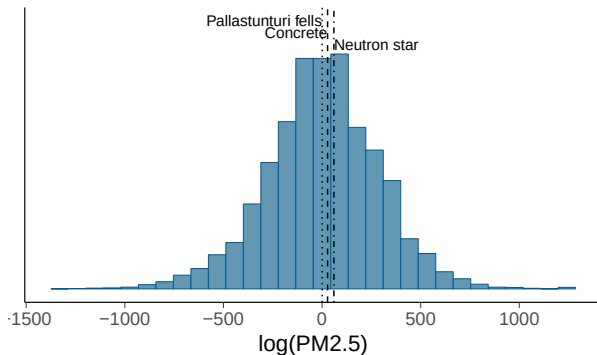


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Example of weakly informative prior

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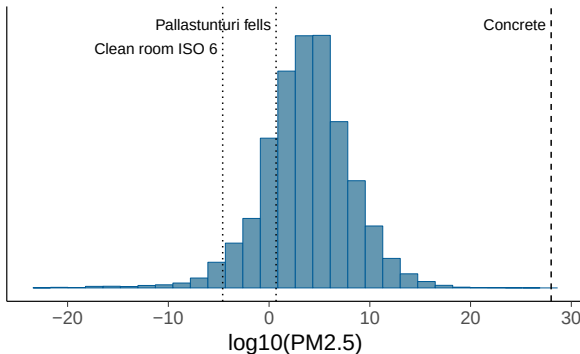


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Example of weakly informative prior

Prior predictive distribution with weakly informative





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Effect of incorrect priors?

- Posterior distributions
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- Introduce bias, but often still produce smaller estimation error because the variance is reduced
 - bias-variance tradeoff



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Sufficient statistics

- The function $t(y)$ of data y is said to be a *sufficient statistic* for θ if the likelihood for θ depends on the data y only through the value of $t(y)$.
- Example: Binomial model (with known n , and $y_i \in \{0, 1\}$)



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after normalization

$$p(\theta|y, n, M) = \text{Beta}(\theta|\alpha + \sum y_i, \beta + n - \sum y_i)$$



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after normalization

$$p(\theta|y, n, M) = \text{Beta}(\theta|\alpha + \sum y_i, \beta + n - \sum y_i)$$

Hence, $\sum y_i$ is a sufficient statistic for θ **in this model**.



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Demo in R

- Posterior distributions
- Predictive distributions
- Prior distributions
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- L2demo.R

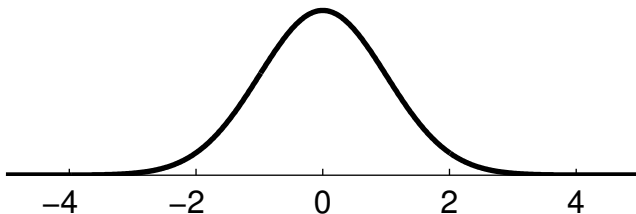


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Normal / Gaussian

- Observations $y \in \mathcal{R}$ (real valued)
- Mean θ and variance σ^2 (or deviation σ)
- For now: assume σ^2 is known

$$p(y|\theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y - \theta)^2\right)$$
$$y \sim \mathcal{N}(\theta, \sigma^2)$$





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Why use the Normal distribution?

- Posterior distributions
- Predictive distributions
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- Many measurements are approximately continuous and symmetric
- Sums and averages are often approximately Normal
 - this is the central limit theorem intuition



Why use the Normal distribution?

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- Many measurements are approximately continuous and symmetric
- Sums and averages are often approximately Normal
 - this is the central limit theorem intuition
- Normal models are mathematically and computationally convenient
- Convenience is not a model check



- Posterior distributions
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Why use the Normal distribution?

- Many measurements are approximately continuous and symmetric
- Sums and averages are often approximately Normal
 - this is the central limit theorem intuition
- Normal models are mathematically and computationally convenient
- Convenience is not a model check
- Always ask:
 - Is the outcome continuous?
 - Is the distribution roughly symmetric?
 - Are extreme values reasonably described by Normal tails?



- Posterior distributions
- Predictive distributions
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Central limit theorem: useful but limited

- Under certain conditions, sums and averages of many random variables become approximately Normal

$$\bar{y} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$



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 - it does not apply to all distributions, for example Cauchy
 - the approximation may be poor for small n
 - it may be poor for binomial data when θ is close to 0 or 1



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- The central limit theorem has limits:
 - it does not apply to all distributions, for example Cauchy
 - the approximation may be poor for small n
 - it may be poor for binomial data when θ is close to 0 or 1
- Discussion question: Are we modeling the raw data, or an average/estimate?



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Normal distribution - conjugate prior for θ

- Assume σ^2 known

Likelihood

$$p(y|\theta) \propto \exp\left(-\frac{1}{2\sigma^2}(y - \theta)^2\right)$$

Prior

$$p(\theta) \propto \exp\left(-\frac{1}{2\tau_0^2}(\theta - \mu_0)^2\right)$$



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$$\exp(a) \exp(b) = \exp(a + b)$$



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Posterior

$$p(\theta|y) \propto \exp\left(-\frac{1}{2}\left[\frac{(y - \theta)^2}{\sigma^2} + \frac{(\theta - \mu_0)^2}{\tau_0^2}\right]\right)$$



- Posterior distributions
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Normal distribution - conjugate prior for θ

- Posterior (see ex 2.14a)

$$\begin{aligned} p(\theta|y) &\propto \exp\left(-\frac{1}{2}\left[\frac{(y-\theta)^2}{\sigma^2} + \frac{(\theta-\mu_0)^2}{\tau_0^2}\right]\right) \\ &\propto \exp\left(-\frac{1}{2\tau_1^2}(\theta-\mu_1)^2\right) \end{aligned}$$

$\theta|y \sim \mathcal{N}(\mu_1, \tau_1^2)$, where

$$\mu_1 = \frac{\frac{1}{\tau_0^2}\mu_0 + \frac{1}{\sigma^2}y}{\frac{1}{\tau_0^2} + \frac{1}{\sigma^2}} \text{ and } \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2}$$



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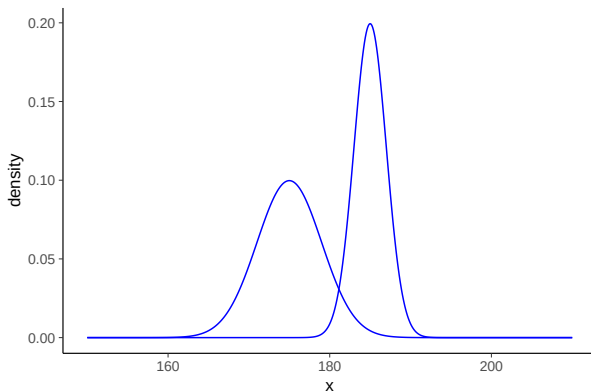
- 1/variance = precision
- Posterior precision = prior precision + data precision
- Posterior mean is precision weighted mean



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- Posterior distributions
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Normal distribution - example

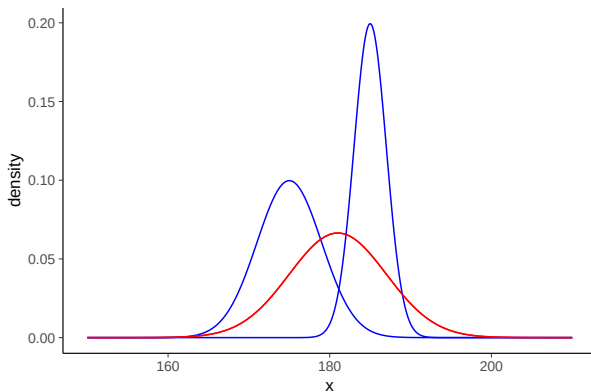


Blue: data point(s), Red: Prior, Purple: Posterior



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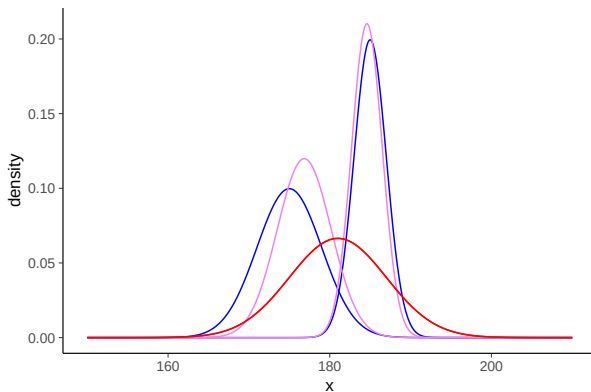


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Normal distribution - conjugate prior for θ

Posterior (several observations $y = (y_1, \dots, y_n)$)

$$p(\theta|y) \propto p(\theta)p(y|\theta)$$



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Normal distribution - conjugate prior for θ

- Assume

$$y_i|\theta \sim \mathcal{N}(\theta, \sigma^2), \quad \theta \sim \mathcal{N}(\mu_0, \tau_0^2),$$

where σ^2 is known



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Normal distribution - conjugate prior for θ

- Assume

$$y_i|\theta \sim \mathcal{N}(\theta, \sigma^2), \quad \theta \sim \mathcal{N}(\mu_0, \tau_0^2),$$

where σ^2 is known

- The posterior is also Normal:

$$\theta|y \sim \mathcal{N}(\mu_n, \tau_n^2)$$



- Posterior distributions
- Predictive distributions
- Prior distributions
- The Normal model
- Poisson Model

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- The posterior mean is a weighted average:

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- Prior mean and data mean are weighted by precision



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Normal distribution - posterior uncertainty

- The posterior precision is

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$$



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- Interpretation:
 - smaller τ_0^2 means a stronger prior
 - smaller σ^2 means more informative observations
 - larger n means more information from data



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- **Discussion question:** What is the sufficient statistic for θ here?



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Normal distribution - limiting cases

- If $\tau_0^2 = \sigma^2$, the prior has the same weight as one observation:

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- If $n \rightarrow \infty$, the data dominate the prior:

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Normal distribution - conjugate prior for θ

- Posterior predictive distribution

$$p(\tilde{y}|y) = \int p(\tilde{y}|\theta)p(\theta|y)d\theta$$

$$p(\tilde{y}|y) \propto \int \exp\left(-\frac{1}{2\sigma^2}(\tilde{y} - \theta)^2\right) \exp\left(-\frac{1}{2\tau_1^2}(\theta - \mu_1)^2\right) d\theta$$

$$\tilde{y}|y \sim \mathcal{N}(\mu_1, \sigma^2 + \tau_1^2)$$



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- Can be derived in multiple ways
 1. integrate
 2. $p(\tilde{y}, \theta)$ is a bivariate normal - marginalize out θ
- Predictive variance
 1. observation model variance σ^2
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- **Discussion question:** Where do aleatoric and epistemic uncertainty appear in $\sigma^2 + \tau_1^2$?



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Poisson model

- Posterior distributions
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- Poisson likelihood/model: $Po(\lambda)$
- Used for **count** data



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- Used for **count** data
- Let $X \sim Po(\lambda)$, then

$$E(X) = V(X) = \lambda$$

- **Discussion:** What would make this assumption questionable for observed count data?



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- **Today:** Mostly the negative binomial is used for count data



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Poisson-Gamma model

- Observations $y = (y_1, \dots, y_n)$

$$y_i | \lambda \sim \text{Poisson}(\lambda),$$
$$\lambda \sim \text{Gamma}(\alpha, \beta)$$

where β is the rate parameter



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$$p(\lambda | y) \propto \lambda^{\sum_i y_i} \exp(-n\lambda) \lambda^{\alpha-1} \exp(-\beta\lambda)$$
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- Thus

$$\lambda | y \sim \text{Gamma} \left(\alpha + \sum_i y_i, \beta + n \right)$$



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Poisson-Gamma posterior predictive

- Integrate over uncertainty in λ

$$p(\tilde{y}|y) = \int p(\tilde{y}|\lambda)p(\lambda|y)d\lambda$$



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- If $\lambda|y \sim \text{Gamma}(\alpha_n, \beta_n)$, then

$$\tilde{y}|y \sim \text{NegBinomial} \left(\text{size} = \alpha_n, \text{prob} = \frac{\beta_n}{\beta_n + 1} \right)$$



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- Here

$$\alpha_n = \alpha + \sum_i y_i, \quad \beta_n = \beta + n$$

- This is the count-data analogue of posterior predictive averaging